MAIS PAC

IPhO 02.01: Kinematics: Describing Motion · Lecture + Problems

Kinematics: Describing Motion

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. Where kinematics starts

Kinematics is the vocabulary of motion before any force appears. Everything follows from one idea: position is a vector, and velocity is its rate of change. In one dimension,

$$x(t) \quad v=(dx)/(dt) \quad a=(dv)/(dt)=(d^2x)/(dt^2). \tag{1}$$

Working backwards you integrate, and each integration brings one constant that a measurement must fix:

$$v = v_0 + at, \quad x = x_0 + v_0t + (1)/(2)\, at^2. \tag{2}$$

Those two lines are **only** true when a is constant. Olympiad problems constantly test whether you know that. If a varies, you must either integrate properly, or read the motion from a graph, or use the energy relation v,dv = a,dx obtained by eliminating time.

2. Reading a graph like a physicist

A velocity–time graph is a complete description of one-dimensional motion.

- **Slope** of v(t) is the acceleration.
- **Signed area** between the curve and the time axis is the displacement.
- **Total path length** uses absolute values of the areas — turning points matter.
- An area **below** the axis is negative displacement, but it is still distance travelled.

An acceleration–time graph gives velocity through the **change** in area: you must know v(0) to locate the curve. A position–time graph read backwards gives velocity by tangents; its curvature tells you the sign of the acceleration.

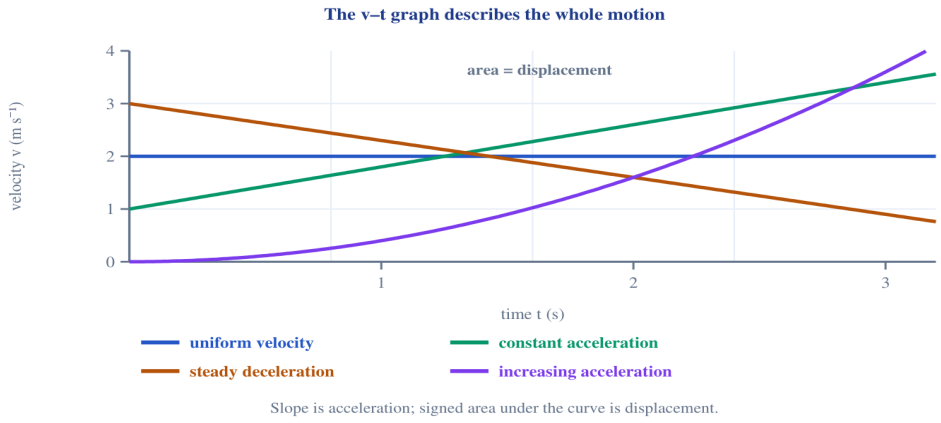


Figure. Slope is acceleration; signed area under the curve is displacement.

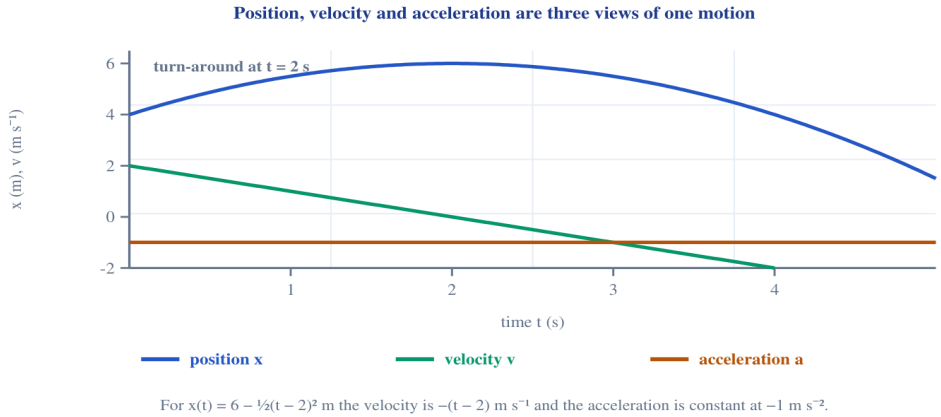


Figure. For x(t) = 6 – ½(t – 2)² m the velocity is –(t – 2) m s■¹ and the acceleration is constant at –1 m s■².

Averaged quantities deserve care. Average velocity is displacement over time, v = Δ x/Δ t, not the average of the speeds. Only for constant acceleration is it also the midpoint value (1)/(2)(v_0+v), which is why that formula appears in every textbook and fails in every other case.

3. Projectile motion: two independent one-dimensional problems

With constant gravity and no air resistance, the axes decouple:

$$x = v_0\cos\theta;\, t,\; y = v_0\sin\theta;\, t - (1)/(2)\, gt^2. \tag{3}$$

Eliminating t gives a parabola,

$$y = x\tan\theta - (gx^2)/(2v_0^2\cos^2\theta). \quad (4)$$

Three results are worth having ready:

- **Time of flight** over level ground: $T = 2v_0\sin\theta/g$.
- **Range**: $R = v_0^2\sin 2\theta/g$, largest at 45°, and equal for complementary angles.
- **Maximum height**: $H = v_0^2\sin^2\theta/(2g)$.

The symmetry that produces “complementary angles share a range” is worth noticing: it is a symmetry of the product $\sin\theta\cos\theta$, not of the trajectory shape.

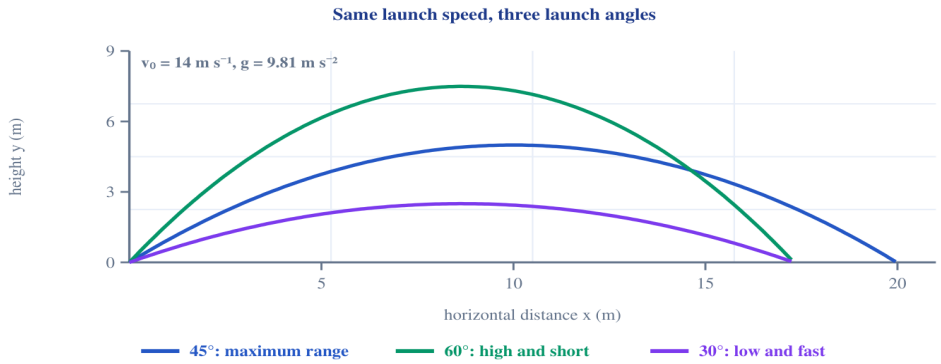


Figure. Complementary angles reach the same range; 45° maximizes it when air resistance is negligible.

When the target is not at the same height, do not reach for the range formula. Write the trajectory equation, set $y = h$, and solve the quadratic for x; keep only the physically meaningful root, and remember that two roots can be a genuine pair of launch angles.

4. Relative motion and frame changes

Velocities add as vectors:

$$v_{A\{\text{ rel ground}\}} = v_{A\{\text{ rel }}B) + v_{B\{\text{ rel ground}\}}. \quad (5)$$

Galilean addition is exact for speeds much less than c; relativity replaces it later. In practice you use it as a **choice**: solve the problem in whichever frame makes the motion trivial, then transform the answer back.

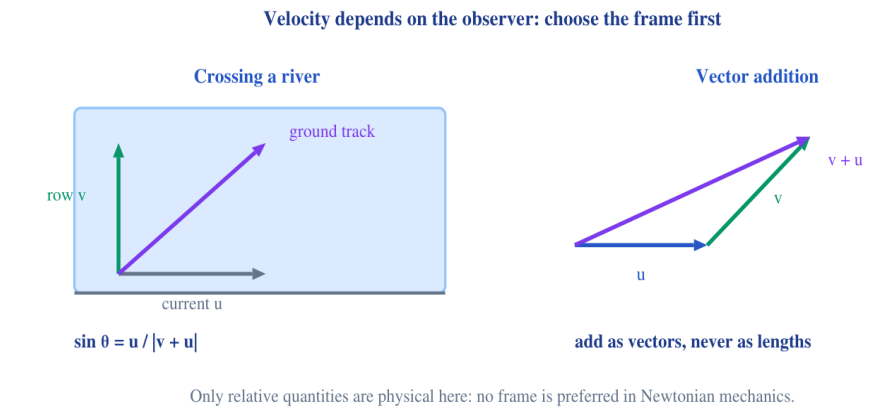


Figure 3. Switch to the frame in which the motion is simplest, solve, then transform back. That single habit solves most “two moving objects” problems.

Two standard traps:

- **Crossing a current.** Minimum crossing time comes from aiming directly across (the current only carries you downstream); arriving directly opposite requires aiming upstream, and if the current is faster than your rowing speed it is impossible.
- **Closing speed.** Two objects approaching each other each at speed v close at 2v in the ground frame but at v in each other's frame. The collision outcome depends on relative speed, and that is a physics statement, not a bookkeeping one.



Figure. A ball dropped from 5 m; air resistance is neglected.

Interactive experiment: Range, apex and time of flight — open the online handout to use it.

5. Choosing coordinates, and the limits of the model

Choose the origin and positive direction once, write it down, and never change it mid-solution. A common mark-losing error is mixing a sign convention halfway.

Three modelling decisions appear in almost every kinematics problem:

1. **Is air resistance negligible?** It is not, for light objects over long falls, or when the question mentions terminal velocity, raindrops or feathers. Including linear drag f=-kv turns constant acceleration into an exponential approach to terminal velocity,

$$v(t)=v_T(1-e^{-t/\tau}),\ v_T=(mg)/(k),\ \tau=(m)/(k). \tag{6}$$

2. **Is the ground flat?** A range formula on a slope needs the intersection of a parabola and a line.
3. **Is the projectile a point?** Spin, shape and a finite size matter when the object is a ball, a disc or a balloon.

The method in one paragraph. Write the coordinate choice. Draw the v–t sketch. Use slope for acceleration and area for displacement. Keep the axes separate for projectiles. Change frame when it simplifies the question. Check the answer in a limit — g→0, θ→0, v_0→∞ — before you believe it.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — From a v–t graph to real motion. A cart moves along a straight track. Its velocity is +4.0 {m,s^(-1)} for 2.0 {s}, then falls linearly to -2.0 {m,s^(-1)} over the next 3.0 {s}. (a) Find the displacement over the whole 5.0 {s}. (b) Find the total distance travelled. (c) Find the acceleration during the second stage, and the instant when the cart turns around.

Worked solution →

[1] Problem 2 — Two stones, one clock. A stone is dropped from rest at the top of a well and a second stone is thrown downward from the same point 1.0 {s} later with speed 12 {m,s^(-1)}. Take g=9.8 {m,s^(-2)}. (a) At what time after the first stone is released do the stones have the same speed? (b) How far below the release point are they then? (c) Explain why the stones can never have the same position while the second is still moving faster.

Worked solution →

[2] Problem 3 — Range on level ground and on a wall. A ball is launched at 20 {m,s^(-1)}. Take g=9.8 {m,s^(-2)} and ignore air resistance. (a) Find the maximum range and the angle that achieves it. (b) Find the two launch angles that put the ball at the same point 25 {m} away on level ground. (c) A wall of height 6.0 {m} stands 20 {m} from the launch point. Which launch angles clear it?

Worked solution →

[2] Problem 4 — The river crossing. A rower can move at 1.6 {m,s^(-1)} relative to the water. The river is 80 {m} wide and flows at 1.2 {m,s^(-1)}. (a) Find the heading that gets the rower straight across, and the time taken. (b) Find the heading that crosses in the shortest time, and how far downstream the rower lands. (c) Show that a straight crossing is impossible if the current exceeds the rowing speed, and describe the two equal-speed boundary case.

Worked solution →

[2] Problem 5 — Projectile over a wall: the algebraic route. A projectile is launched from ground level at speed v_0 and angle θ. A vertical wall of height H stands at horizontal distance d. (a) Derive the condition on v_0 for the projectile to pass over the wall. (b) Show that for a fixed speed there is an optimal angle and find it. (c) Show that, for a target at height y and distance x, a launch at speed v_0 is possible only if $v_0^2 \geq g(y + \sqrt{(x^2+y^2)})$, and interpret the second term.

Worked solution →

[2] Problem 6 — Reading an acceleration graph. An object starts from rest at x=0. Its acceleration is +2.0 {m,s^(-2)} for 3.0 {s}, zero for 2.0 {s}, then -3.0 {m,s^(-2)} until the object comes to rest. (a) Sketch a(t), v(t) and x(t) and label their shapes. (b) Find the maximum speed and when it occurs. (c) Find the total time and the final position.

Worked solution →

[2] Problem 7 — Drag changes everything. A raindrop of mass m falls from rest under gravity with linear drag f = -kv. (a) Show that the terminal speed is v_T = mg/k and find the time constant τ = m/k. (b) Show that the distance fallen by time t is v_T[t - τ(1-e^(-t/τ))]. (c) Evaluate how far the drop has fallen at t=3τ as a fraction of v_Tt, and explain the difference from the drag-free case.

Worked solution →

[2] Problem 8 — Two-dimensional chase. At t=0 a boat is at the origin and moves with constant velocity (3.0, 0) {m,s^(-1)}. A second boat starts at (0, 40) {m} and moves with constant velocity (0, -5.0) {m,s^(-1)}. (a) Find the separation as a function of time. (b) Find the minimum separation and when it occurs. (c) A third boat must leave the origin at t=0 and intercept the second boat at the moment of minimum separation: find its required velocity.

Worked solution →

[2] Problem 9 — Throwing from a cliff. A ball is thrown from a cliff of height h at speed v_0 and angle θ above the horizontal. (a) Find the time of flight from the quadratic in t, and interpret the negative root. (b) Show that the horizontal range is R = (v_0cos θ)/(g)(v_0sin θ + √(v_0²sin ^2θ + 2gh)). (c) Show that as h→∞ the optimal angle tends to 0, and explain this physically.

Worked solution →

[3] Problem 10 — A rotating reference frame. A turntable rotates with angular velocity ω. A bead slides frictionlessly in a straight radial groove. (a) Write the radial acceleration in the rotating frame and identify the two fictitious terms. (b) Show that the bead's radial position obeys r = ω² r and hence r = r_0 ω t if it starts at rest at radius r_0. (c) Explain why a bead released from rest in the inertial frame does **not** stay on the groove, and calculate the sideways force the groove must supply. (d) Show that the result reduces to r ≈ 0 for short times t≪ 1/ω.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

HyperPhysics mechanics map

A compact concept map for revising definitions and standard results quickly.